

System Design (Telusko) - Day 9

One-Shot 5h course | Video 2:00-2:15

Indexing

Speed up reads with database indexes.

Key concepts

- An index is a sorted structure (often B-tree) for fast lookups.
- Speeds reads but slows writes and uses storage.
- Composite indexes for multi-column queries.
- Index the columns you filter/sort/join on.

Key example

WHERE email = ? -> index on email turns a full scan into a fast lookup.

Today's tasks (1.5h)

- ☐ Learn B-tree indexes
- ☐ Add an index to a slow query
- ☐ Note the write trade-off

Resources

Video: youtu.be/Vnm-ycSfJx4 (start at 2:00-2:15)

Docs: docs.telusko.com/docs/system-design

(Full clickable links are in the app: Goals > this day > Study block.)